

MEZEN, Russian Federation

WMO: 224710

Lat: 65.8725N

Lon: 44.2081E

Elev: 62

StdP: 14.66

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -27.2	(c) -21.5	(d) -31.8	(e) 0.9	(f) -27.0	(g) -25.9	(h) 1.3	(i) -21.2	(j) 27.2	(k) 19.5	(l) 23.3	(m) 18.4	(n) 3.3	(o) 180	(p) 0.755

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 16.4	(c) 78.3	(d) 64.8	(e) 74.2	(f) 62.4	(g) 70.6	(h) 60.9	(i) 66.7	(j) 75.1	(k) 64.4	(l) 72.4	(m) 62.1	(n) 68.5	(o) 8.9	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.3	87.2	71.1	61.1	80.8	68.3	59.0	74.7	65.8	31.3	75.3	29.6	72.9	28.0	68.4	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 22.6	(b) 20.3	(c) 18.0	(e) -30.9	(f) 84.2	(g) 5.5	(h) 3.2	(i) -34.9	(j) 86.6	(k) -38.1	(l) 88.4	(m) -41.2	(n) 90.3	(o) -45.2	(p) 92.6		
(a) 22.6	(b) 20.3	(c) 18.0	(e) -31.1	(f) 69.9	(g) 5.5	(h) 3.0	(i) -35.0	(j) 72.0	(k) -38.2	(l) 73.8	(m) -41.3	(n) 75.5	(o) -45.2	(p) 77.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	32.7	8.5	9.1	19.9	30.3	41.4	51.5	58.0	54.4	46.5	34.8	22.2	14.6	
	DBStd	20.07	14.44	16.18	10.52	8.64	9.99	8.20	7.99	6.82	6.54	6.94	11.90	14.18	
	HDD50	6941	1287	1144	934	593	307	80	20	29	144	470	835	1098	
	HDD65	11816	1752	1564	1399	1042	731	411	241	337	556	935	1285	1563	
	CDD50	639	0	0	0	0	41	125	270	165	38	0	0	0	
	CDD65	39	0	0	0	0	1	5	25	8	0	0	0	0	
	CDH74	355	0	0	0	0	16	58	225	56	0	0	0	0	
	CDH80	54	0	0	0	0	2	8	38	6	0	0	0	0	
Wind	WSAvg	8.7	8.9	8.8	9.0	9.1	9.6	9.0	8.0	7.8	7.9	8.9	8.8	9.1	
Precipitation	PrecAvg	22.0	1.3	1.0	1.1	1.0	1.6	2.2	2.5	3.1	2.6	2.3	1.7	1.6	
	PrecMax	28.7	2.1	2.1	2.5	1.9	4.1	5.4	7.1	6.3	5.2	3.4	3.4	2.7	
	PrecMin	18.2	0.6	0.4	0.3	0.1	0.4	0.6	0.2	0.5	0.8	1.2	0.7	0.6	
	PrecStd	2.7	0.4	0.4	0.6	0.5	0.8	1.0	1.7	1.2	1.1	0.6	0.7	0.5	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.1	36.0	40.7	56.1	76.1	80.1	84.0	80.3	68.2	53.7	40.8	35.9	
		MCWB	32.9	34.6	35.4	45.6	59.7	66.1	68.4	66.8	60.6	50.1	40.0	34.2	
	2%	DB	32.3	34.1	36.9	48.5	68.9	74.7	79.8	74.1	63.6	48.9	37.8	34.2	
		MCWB	31.2	32.8	33.3	41.3	55.2	60.1	65.4	63.9	56.8	46.3	37.0	33.0	
	5%	DB	29.8	32.4	35.1	44.1	63.6	70.7	75.8	69.6	59.8	45.9	35.8	33.1	
		MCWB	28.8	31.4	32.6	38.5	52.1	59.0	64.4	61.7	54.6	43.9	34.7	32.0	
	10%	DB	26.7	30.0	33.4	40.8	58.4	66.2	72.3	66.0	56.5	43.5	34.5	31.4	
		MCWB	25.7	29.0	31.3	36.2	49.2	56.8	62.6	59.7	52.8	41.8	33.5	30.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.3	35.0	36.0	46.3	61.1	67.4	71.4	69.0	61.6	51.5	40.1	34.6
			MCDB	33.8	35.7	39.0	54.6	71.3	77.6	81.6	77.7	67.0	53.5	40.5	35.1
2%		WB	31.5	33.1	34.3	42.2	57.0	63.6	67.4	65.2	58.0	46.6	37.3	33.5	
		MCDB	32.4	33.8	36.2	47.5	66.3	70.9	75.6	71.8	62.1	48.3	37.7	34.3	
5%		WB	28.8	31.6	33.0	39.1	53.3	60.5	65.5	62.6	55.4	44.1	35.0	32.1	
		MCDB	29.7	32.5	34.8	43.0	61.8	67.5	74.0	68.4	58.8	45.7	35.5	33.1	
10%		WB	25.8	28.9	31.5	37.0	49.6	57.7	63.5	60.3	53.0	42.1	33.6	30.4	
		MCDB	26.7	30.1	33.4	40.2	58.0	65.0	71.3	65.1	56.1	43.4	34.3	31.3	
Mean Daily Temperature Range		5% DB	MDBR	11.6	11.4	13.4	12.8	15.1	15.9	16.4	14.0	10.9	7.1	7.8	10.3
			MCDBR	9.3	6.3	10.5	15.8	24.5	23.0	22.6	19.4	16.5	9.5	5.9	6.9
	MCWBR		9.1	6.2	8.0	10.4	13.6	12.2	11.8	11.3	10.3	7.6	5.8	6.6	
	5% WB	MCDBR	9.3	5.6	8.5	14.1	23.2	19.8	20.2	17.2	13.7	9.2	5.9	6.6	
		MCWBR	9.1	5.6	6.9	9.8	13.9	12.4	12.0	11.2	9.5	7.8	6.1	6.6	
Clear-Sky Solar Irradiance	taub	0.217	0.274	0.306	0.338	0.345	0.345	0.382	0.355	0.328	0.299	0.359	0.218		
	taud	1.924	2.061	2.113	2.151	2.400	2.480	2.392	2.444	2.499	2.313	2.579	1.943		
	Ebn at Noon	69	180	235	259	272	275	259	255	237	177	70	9		
	Edh at Noon	6	19	29	36	31	30	32	27	21	15	6	2		
All-Sky Solar Radiation	RadAvg	30	196	657	1201	1503	1637	1575	1061	570	201	43	5		
	RadStd	3	28	73	114	129	166	174	115	57	35	6	0		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page